WEEK 4 · LESSON 24 OF 24 · HANDOUT

Final Project: A Deep-Learning System and Certificate

Study time: **about 120 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=ZAGiinWiFsE — open the lesson online to watch inside the course.

🎓 **Go deeper:** freeCodeCamp: PyTorch for Deep Learning & Machine Learning — full course — https://www.youtube.com/watch?v=V_xro1bcAuA

🎓 **Go deeper:** MIT 6.S191: Introduction to Deep Learning (2024) — https://www.youtube.com/watch?v=ErnWZxJovaM

Your goal in this lesson: Design, build, evaluate and present a complete deep-learning system for a real problem, using everything in this course — and earn your certificate.

Choose your track

Track	Problem	Starting point
A — Vision	Recognise 3+ classes of images (crop conditions, local fruits, waste types, road signs…)	Week 3 project: ImageFolder + pretrained ResNet
B — Forecasting	Forecast a daily or hourly series (sales, electricity, rainfall, clinic visits)	Lesson 20 LSTM, with seasonal baselines
C — Language	Sort messages, reviews or questions into categories, in English, Kinyarwanda or both	Lessons 19 and 21: embeddings and a transformer

Use **real data** where you can: your own photos (with permission), open data from the National Institute of Statistics of Rwanda or Kaggle, or anonymised data from your workplace. If you cannot get real data, extend the simulated datasets from this course and say so clearly.

The seven parts (every track)

1. **Problem and users** — who will use it, what decision it supports, what a mistake costs.
2. **Data** — source, size, classes or variables, cleaning, train / validation / test split (by time for forecasting), consent and privacy.
3. **Baseline** — a simple method first: logistic regression, "same as last week", or a bag of embeddings.
4. **Deep model** — architecture, why you chose it, augmentation or regularisation, optimiser, learning rate, early stopping.
5. **Evaluation** — test results used once, compared with the baseline; confusion matrix or error plot; mistakes studied; results by group where relevant.
6. **Trust** — confidence rule for human review; saliency maps (vision) or examples of hard cases; limits.
7. **Deployment plan** — checkpoint, a FastAPI endpoint (tested with `TestClient`), monitoring, retraining, model card.

Starter checklist (copy into your notebook)

```
CHECKLIST = [
    "Problem, users and cost of mistakes written down",
    "Data source, licence and consent recorded",
    "Train / validation / test split made before any modelling",
    "Baseline trained and scored",
    "Deep model trained with early stopping on validation data",
    "Test set used exactly once",
    "Mistakes and confidence studied",
    "Checkpoint saved with its settings",
    "API endpoint tested with TestClient",
    "Model card and monitoring plan written",
]
done = {item: False for item in CHECKLIST}
done["Problem, users and cost of mistakes written down"] = True
print(f"{sum(done.values())} of {len(done)} steps done")
for item, ok in done.items():
    print("[x]" if ok else "[ ]", item)
```

Present it (10 minutes)

- Slide 1: the problem and the users.
- Slide 2: the data and its limits.
- Slide 3: baseline vs deep model (one table).
- Slide 4: one chart (learning curves, confusion matrix or forecast plot).
- Slide 5: risks, human oversight and the deployment plan.

Marking guide (self-check)

Area	Excellent work
Problem	Real users and costs of mistakes are clear
Data	Honest description, good splits, privacy respected
Method	Baseline first; deep model trained with validation and early stopping
Evaluation	Test used once; mistakes and groups analysed
Trust and deployment	Confidence rule, tested API, monitoring plan, model card
Communication	A non-expert understands the result in 10 minutes

Getting your certificate

1. Complete **every lesson** in the course.
2. The **final exam** then unlocks on the course page: 60 questions (single answer, multiple answers, true/false and fill in the blank), 90 minutes, 60% to pass, 3 attempts. The exam opens in full screen; leaving the exam screen is counted, and the third time ends the exam automatically.
3. Pass the exam and your **WAMUGII Certificate** appears on the course page with your score. Add it — and your project notebook — to your CV and LinkedIn.

What you can do now

You can build neural networks from scratch and with PyTorch; train them well with good optimisers, regularisation, normalisation and early stopping; build CNNs with augmentation and transfer learning; work with sequences using embeddings, LSTMs and attention; create autoencoders; and deploy models responsibly. Next in the WAMUGII AI Program: **Modern AI — Large Language Models and Generative AI**, where you will use and adapt the transformers behind today's AI assistants.

Remember this: A deep-learning system is more than a network: real users, honest data, a baseline, careful evaluation, human oversight and a deployment plan make it worth using.

Name: _____ Date: _____

Activity

Task 1

Complete the seven parts for your track and present your project. Ask your instructor for feedback in the lesson Q&A below — share your notebook link and one question you want help with.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What should you build before the deep model?
   A. The API
   B. Nothing
   C. A simple baseline to compare against
   D. The certificate
2. How must a time-series project split its data?
   A. By label
   B. Randomly
   C. It does not matter
   D. By time: train on the past, test on the future
3. How many times should the test set be used?
   A. Every epoch
   B. Twice
   C. Once, at the end
   D. Never
4. Which belongs in the trust part of the project?
   A. Hiding mistakes
   B. Using the test set for tuning
   C. Removing the baseline
   D. A confidence rule that sends unsure cases to a person
5. What is required before the final exam unlocks?
   A. Passing a quiz once
   B. Nothing
   C. Paying a fee
   D. Completing every lesson
6. What is the pass mark of the final exam?
   A. 50%
   B. 75%
   C. 100%
   D. 60%

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